Document Control

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Exchange transfusion Guideline for Neonates			
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1. Purpose

1.1. The purpose of this document is to detail the process for exchange transfusion for neonates.

1.2. The policy applies to all neonatal staff involved in the care of neonates requiring exchange transfusion at North Devon District Hospital.

1.3. Implementation of this policy will ensure that:

- The procedure is completed as safely as possible
- There is minimal distress caused to the neonate
- We practice in accordance with the best practice guidance for exchange transfusion for neonates

2. Responsibilities

Role of Neonatal Nurses

2.1. The neonatal nurse is responsible for:

- Competently assisting the clinician in the procedure of exchange transfusion
- Ensuring that all necessary equipment is available

- Ensuring that the blood required is available and ready

Role of Medical staff

2.2. The Medical staff are responsible for:

- Liaising with the tertiary unit to ensure an agreed plan of care for the patient, including the possibility the infant may need to be transferred
- The clinician should be familiar with the procedure used and comfortable to assist where required.

3. Background

3.1. "Exchange transfusion must take place in an intensive care setting with intensive physiological and biochemical monitoring, carried out by staff trained in the procedure following written informed parental consent" (BCSH, 2016).

3.2. All effort should be made to transfer the at risk baby in/ex utero to an obstetric department with a level 2 or 3 neonatal unit. However due to the rural nature of SCU and the time needed to retrieve a baby to a NICU (usually more than 6 hours) this guidance has been written. Exchange transfusion should only occur in SCU in life threatening circumstances following instruction and consultation with the tertiary unit.

3.3. Neonatal red cell exchange transfusion is mainly used in the treatment of severe hyperbilirubinemia or anaemia in babies with haemolytic disease of the fetus or newborn (HDFN). It removes antibody-coated neonatal red cells and reduces the level of plasma unconjugated bilirubin (the cause of bilirubin encephalopathy). The steep decline in the incidence of HDFN following the introduction of maternal anti-D Ig prophylaxis, more effective antenatal monitoring and treatment, and the use of intensive phototherapy and intravenous immunoglobulin postnatally has made red cell exchange transfusion an uncommon procedure that should only be performed in specialist units by experienced staff (Joint United Kingdom Blood Transfusion and Tissue Transplantation Services Professional Advisory Committee (JPAC), 2021).

3.4. The aim of the exchange transfusion is to decrease the rate of haemolysis by reducing the amount of circulating anti-foetal blood cell antibodies and also reduce the level of circulating bilirubin and thus reduce the risk of kernicterus (Behrsin, et.al., 2019).

3.5. Intravenous immunoglobulin (IVIG) (500 mg/kg over 4 hours) is used as an adjunct to continuous intensified phototherapy in cases of rhesus haemolytic disease or ABO haemolytic disease when the serum bilirubin continues to rise by more than 8.5 micromol/litre per hour (NICE, 2021).

3.6. 'Partial exchange transfusion' can be used for the management of polycythaemia, although this treatment remains controversial. A systematic review found that crystalloid solutions are as effective as colloid solutions for partial exchange transfusion in newborn babies with polycythaemia. The authors' conclusions are in line with the evidence presented, but should be treated with caution given that they were based on a few small studies. To minimise risks associated with use of blood products, normal saline should be used if haemodilution is undertaken.

4. Indication for exchange transfusion in hyperbilirubineamia (NICE 2021)

4.1. Infants whose serum bilirubin level indicates its necessity (see NICE threshold tables and the treatment threshold graphs from the CG98 full guideline)

4.2. and/ or with clinical features and signs of acute bilirubin encephalopathy.

5. Procedure for exchange transfusion

Steps	Preparation for Exchange Transfusion
1	Frequent consultant to consultant liaison with tertiary unit. With an expectation that the Peninsula Neonatal Transport service will be coming as soon as possible to take over the care of this infant.
2	**Parent/carer Information and consent** Informed parental consent should be obtained and documented wherever possible this should be written consent. NICE (2021) recommend information about exchange transfusion and intravenous immunoglobulin should include: • the fact that exchange transfusion requires that the baby be admitted to an intensive care bed • why an exchange transfusion is being considered • why an exchange transfusion may be needed to treat significant hyperbilirubinaemia • the possible adverse effects of exchange transfusions • when it will be possible for parents or carers to see and hold the baby after the exchange transfusion Follow Trust policy for Patients Refusing a Blood Transfusion when parents/ carers refuse to give consent.
3	Initial investigations should include: • FBC, U&E's, LFT'S, Calcium (Ca2+) • Blood glucose

	• Arterial blood gas • Maternal crossmatch sample for blood group, antibody screen and antiglobulin test • Baby blood group ABO and Rh(D) type and Direct Antiglobulin Test (DAT). Also test for antigen to any specific antibody found in maternal blood. • Maternal blood group (ABO and Rh(D) type) and antibody screen / identification Order red blood cells (follow guidelines for red blood cell transfusion for neonates). Use a double-volume exchange transfusion (NICE, 2021). A 'double volume exchange' (160–200 mL/kg) removes around 90% of neonatal red cells and 50% of bilirubin (JPAC, 2021). **Red Blood Cell Requirements** (JPAC , 2021) Plasma reduced with haematocrit of 0.5–0.6 (NHSBT 0.5–0.55) to reduce the risk of post-exchange polycythaemia In CPD anticoagulant Less than 5 days old Irradiated (essential if previous IUT) CMV negative Sickle screen negative Usually produced as group O (with low-titre haemolysins) RhD negative (or RhD identical with neonate) and Kell negative Red cell antigen negative for maternal alloantibodies IAT crossmatch compatible with maternal plasma
4	**Gain intravenous access** a)UVC & UAC (Optimum) b) Peripheral Cannula & UAC c) Peripheral Cannula x 2 & Peripheral Arterial line d) UVC Only Note arterial lines (umbilical or peripheral should only be used for withdrawal of baby's blood not for the infusion of donor blood)
5	**Consider use of Intravenous Immunoglobulin (IVIG)**
6	**Prepare equipment** • Appropriate means of maintaining the baby's thermoregulation • Appropriate continuous temperature monitoring equipment

	• Cardio-respiratory monitoring equipment, including SpO2, cardiac wave form and blood pressure • Blood sugar monitoring equipment • Trolley • Exchange Transfusion tray • Blood (see below) do not collect blood until ready to begin exchange • Protective clothing - Sterile gowns, gloves and masks • Blood warmer and extension set, (obtained from Delivery Suite, Equipment library or ITU). • Blood giving set with appropriate 170-200 micron filter • Stand for blood • Blood bottles = Full blood count x 2, urea and electrolytes x 2 including calcium and glucose. Blood culture x 2, capillary gas for Ph x 2, (x 2 = pre and post specimens required) • 2 ml, 5 ml, 10 ml and 60 ml Luer lock syringes, needles and a 3 way tap • Resuscitation equipment • Ventilator and intubation equipment, also suction equipment • Sharps bin according to trust policy • High stools for doctor and nurse as the procedure up to 3 hours. • Timing device for timing infusion and withdrawal of aliquots (*see below step 3 Management during exchange transfusion)*
7	Ensure a neonatal screening blood spot has been taken prior to procedure.
8	Call in trained staff to help with procedure and to manage other patients on ward. A minimum of 3 staff are required for the procedure (consultant and 2 trained nurses).
Steps	**Management prior to exchange transfusion**
1	Do not interrupt phototherapy during exchange (NICE, 2021)
2	Nurse the infant in the radiant overhead heater if available, in the intensive care nursery with continuation of phototherapy.
3	Monitor apex, respiratory rate, oxygen saturation, blood pressure and temperature, a baseline set of observations including general colour and tone and blood sugar should be documented before the procedure starts.
4	Stop enteral feeds and aspirate the stomach contents, leaving the naso-gastric tube on free drainage, (to reduce the risk of necrotizing enterocolitis).
5	Recheck SBR and check blood sugar (and effect of treatment) prior to picking up blood
6	Set up the blood warmer following manufacturer's instructions. Use aseptic non touch technique where appropriate.
7	Collect and check the blood with another trained member of staff. See hospital policy on collection of blood. Prime tubing and blood warmer. Do not use albumin priming.

8	Establish volume of aliquots prior to commencement Aliquots usually tolerated for exchange transfusion: Less than 1500g 5ml 1500g – 2499g 10ml 2500g – 3500g 15ml Greater than 3500g 20ml
9	Bloods tests, blood sugar and a starting set of observations are recorded
Steps	**Management during Exchange Transfusion**
1	The infusion may also be given via pump (as used for a normal transfusion). Person A is responsible for infusing the aliquots, (Senior Paediatrician / Consultant) Person B is responsible for withdrawing an equal amount of blood, (Neonatal Nurse). Person C is responsible for the accurate recording of blood input and withdrawal on the Exchange Transfusion Fluid Chart (appendix two) Persons A and B inform person C on completion of each input and withdrawal All clinical staff must be ready to start simultaneously.
2	One member of staff (person A) flushes the input line with sodium chloride and connects the blood giving set, the other staff member (person B) connects the first syringe to the withdrawal line. Start by withdrawal of aliquot. If using a umbilical venous/ arterial line Person C should inspect and document the appearance of toes, fingers and buttocks for signs of circulatory compromise.
3	Person A informs Person B when each aliquot has been infused so that person B can maintain and equal rate of withdrawal. Person C records this. *Small aliquots exchanged at a slower rate are usually better tolerated by infants with cardiovascular instability. As a guide the time for each in/out should be around 4-10 minutes (withdrawal and infusion of blood should be performed at the same rate). In and out aliquots are repeated sequentially until desired volume for exchange is reached. In and out aliquots should be called out, so that Person C recording the procedure can keep an accurate tally.* *A timer may be used for guiding speed of aliquots*
4	Whilst the exchange transfusion is in progress the volume of blood exchanged in and out is recorded in the fluid balance chart with the Time, (see appendix two).
5	The first aliquot of blood removed should be sent for estimation of full blood count, (FBC), differential, urea and electrolytes (U&Es), liver function tests, split bilirubin, Ca, Phosphate, haematocrit and true glucose.

	These will be repeated at regular intervals throughout the procedure In severe cases consider:- coagulation screen, glucose 6-phosphate dehydrogenate (G6 PD).
6	This process of administration and withdrawal continues until the agreed volume of blood has been exchanged or the condition of the baby determines cessation. At the end of the procedure the infant's fluid balance should be in deficit of one aliquot.
7	The final aliquot withdrawn should be sent for blood sugar, urea and electrolytes, bilirubin - total + conjugated, FBC, haematocrit, magnesium, calcium, blood gases and consider coagulation screen.
8	Continuously Observe the Baby • observe the infant constantly for any signs of deterioration including cyanosis, pallor, abdominal distension, vomiting and blood in stools • record apex, respirations, temperature, oxygen saturation and blood pressure every 15 minutes, any changes in these should be reported immediately. • observe infant for signs of hypocalcaemia - Tachycardia, irritability, seizures. Prolonged Q-T interval, (see below section 8.3 for emergency treatment) • monitor capillary blood glucose every 15-30 minutes. • All lines should be monitored for extravasation, blanching, erythema, leaking and oedema, using Neonatal VIP scoring system, with hourly documentation recorded, (Gallant and Schultz, 2006).
9	Aim for procedure should take 2 hours (not longer than 3 hours). On completion of the exchange transfusion, the lines are again flushed with NaCl prior to recommencing the prescribed regime.
10	Disposal of equipment –in accordance with local Trust policy
11	Documentation – the procedure should be contemporaneously documented including: all transfusion records in accordance with national and hospital trust policies and guidelines, timing, accurate fluid balance, observation, samples taken and their results. Also any untoward reaction, corrective measures taken and outcomes.
12	The baby should be left in a comfortable position and parents informed that the procedure has been completed and condition of the baby.
13	Stop the procedure immediately and review if there are any concerns or untoward reactions.
14	Do not routinely administer intravenous calcium (NICE 2021).
Steps	**Management following exchange transfusion**

1	Continue multiple phototherapy if the exchange was for hyperbilirubinaemia, checking serum bilirubin as required.
2	Take blood pressure at completion of procedure. Continue to monitor vital signs hourly for 6 hours or as the condition of the baby indicates. After this time, and if stable and within normal limits routine observations may be recommenced.
3	Measure SBR level within 2 hours and manage according to threshold graphs, (NICE, 2021). Continue to monitor FBC and reticulocytes.
4	Monitor the blood sugar 1,2 and 4 hours after exchange regularly according to condition for rebound hypoglycaemia.
5	Observe catheter sites for bleeding/signs of infection. Umbilical/peripheral arterial line must be removed by a member of staff competent in procedure once lines no longer needed.
6	Test urine for blood, urinalysis, observe stools for blood.
7	Observe for abdominal distension, and for signs of feed intolerance, gastric aspirate and vomiting. Measure abdominal girth 3-4 hourly for 24 hours. Avoid enteral feeds for 24 hours. Re - commence feeds on medical advice,
8	*While these babies still have evidence of haemolysis they should receive folic acid supplementation.(BCSH 2016)*
9	Neonatal screening must not be taken until 4 days after the exchange as blood transfusion will invalidate all the tests.
10	All infants should receive regular audiology testing as they are at risk of hearing loss due to the severely raised serum bilirubin levels
11	Continue to liaise closely with tertiary unit to consider transfer to NICU.

6. Risks/Complications

6.1. Exchange Transfusion Reaction

- Tachycardia or bradycardia.
- Respiratory Distress.
- Dramatic changes in blood pressure.
- Temperature irregularity.
- Rash.
- Cyanosis or pallor.
- Hypoglycaemia
- Hypocalcaemia (treatment see below)
- Cardiac Arrythmias.
- Heart failure (fluid overload).

6.2 Other risk/complications

- Administration of incorrect blood product
- Volume overload

- Acid base instability.
- Apnoea.
- Cardiac arrhythmias.
- Electrolyte imbalances (hypocalcaemia, hypoglycaemia, hypomagnesaemia).
- Convulsions
- Hypotension (volume depletion).
- Embolism (air/thrombus).
- Graft versus host reaction.
- Haemodynamic instability.
- Infection.
- Necrotising enterocolitis.
- Temperature instability.
- Thrombocytopaenia.
- Thrombosis.
- Intraventricular haemorrhage.
- Line complications including line infection and thrombo-embolism
- Coagulopathy

7. Insert next section heading for guideline (if required)

Subsection

7.1. (Insert details if required)

Next subsection

7.2. Details etc…… keep adding sections and subsections as required

8. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

8.1. Key performance indicators comprise:

- Nice – Quality standard for Neonatal Jaundice
- Nice Neonatal Quality Standards
- NHS Toolkit for High Quality Neonatal Services
- National Neonatal Audit Programme
- NHS Standard Contract for Neonatal Critical Care
- ATAIN

Process for Implementation and Monitoring Compliance and Effectiveness

8.2. Staff are informed of new documentation. There is an expectation that staff are responsible to keep updated on any revisions/improvements to practice and deliver care accordingly.

8.3. All Exchange Transfusions will be discussed and monitored for compliance during SCU Governance meetings

8.4. Activity for SCU including intensive care days are monitored using badger data base.

8.5. Intensive care incidents are monitored by the neonatal governance team and neonatal network. All exchange transfusions will be reported by the Datix system and South West Neonatal Network incident reporting process.

8.6. Data is also collected and monitored monthly via the South West Neonatal Network dashboard. Exception reporting occurs monthly via this dashboard and exchange transfusion will be reported as an exception to the South West Neonatal Network. This data including exception reports are compared/discussed during Network Governance meetings. .

8.7. Non-adherence is reviewed and action plans made if required. Discussion and reviews occur at South West Neonatal Network governance meetings, Trust Directorate meetings, Paediatric/SCU Governance meetings and Ward meetings. Learning and action plans are cascaded at these meetings and improvements implemented. Key findings and learning points will be disseminated across network and to relevant staff.

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10. Associated Documentation

- Guidelines for Red Blood Cell Transfusion
- Controlled Storage of Blood and Blood Products Standard Operating Procedure
- Umbilical Catheter Insertion and Management Guidelines
- Arterial Blood Sampling for Neonates Standard Operating Procedure
- Guidelines for Management of Neonatal Jaundice
- Trust policy for Patients Refusing a Blood Transfusion

Appendix One - The use Intravenous immunoglobulin (IVIG) in neonatal haemolytic disease

AAP (1) and NICE (2) recommend the use of IVIG in patients with hyperbilirubinaemia.

Cochrane (3) concluded that these recommendations were based on small studies of poor quality and did not recommend its use.

Since then there have been several more studies and meta-analysis of IVIG in Rhesus and ABO disease for treatment and prophylaxis (after intra-uterine exchange transfusion).
Apart from variation in set up (ABO vs RH disease; prophylactic/treatment; dose and dosing regime,; use and dosage of PTX, level of intervention) they come to different results regarding the effect of IVIG, this not only with regards to avoiding phototherapy. (also LOS, transfusion requirement, rise in SBR level, need for/length of PTX), for references see below.
More recent studies did not show any significant effect. (4, 6).
Most trials did not report any side effects (or absence of these). The only severe complication where reported in two patients with NEC in the IVIG group (no complication in the control) (5).

The use of IVIG cannot generally be recommended and should be made at the discretion of the Consultant Paediatrician on call, guided after consultant to consultant conversation with the neonatologist in the tertiary unit.

(1)American Academy of Pediatrics Subcommittee on Hyperbilirubinemia. Management of hyperbilirubinemia in the newborn infant 35 or more weeks of gestation. Pediatrics. 2004;114(1):297–316
(2) Nice Guidance 98, http://www.nice.org.uk/guidance/cg98/chapter/guidance guidance on IVIG
(3) Immunoglobulin infusion for isoimmune haemolytic jaundice in neonates
Gary S Alcock1,*, Helen Liley2, Published Online: 22 JUL 2002
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Appendix two - Exchange Transfusion Fluid Chart

PAS Label	
Name	
DOB	
Hosp no	
Date	
Blood Group	
Rhesus	
DAT	
Weight	
Gest	
Day	
Feed	
Fluid	
Fluid rate	
Neonatal blood spot	
Pre start blood sugar	
Feed/fluid stopped (time)	
Pre exchange obs	

PAS Label	
Mothers Name	
DOB	
Hosp no	
Blood Group	
Rhesus	
Written consent gained	
Transfusion	mls/kg
Vol to be exchanged	mls
Aliquot size	mls
Aliquot number	
Aliquot frequency	mins

Baby's Blood Test Results pre and during Exchange

Date	Pre Ex	First Aliquot	Time	Time	Time	Time	Time	Time	Time	Time	Time	Time	Time
Hb													
plat													
INR													
APTTr													
SBR													
Na													
K													
Ur													
Crea													
Cal													
pH													
PCO2													
PO2													
HCO3													
B E													
Lactate													
Sugar													
Type													

Exchange Transfusion Fluid Chart

PAS Label
Name
DOB
Hosp no

Aliquot Number	Time	In		Out		Total Balance in-out =	Blood taken (amount +tested for, include in balance)
		Vol aliquot mls	Total mls	Vol aliquot mls	Total mls		
1		XXXXXXX	XXXXXXX				
2							
3							
4							
5							
6							
7							
8							
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